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Genotyping of repetitive elements in WES and WGS data and their validation in DNA samples from PPMI

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Summary

Retrotransposons are highly repetitive elements contributing to 45% of the human genome and a subset of these reference elements are polymorphic for their presence or absence and are known as retrotransposons insertion polymorphisms (RIPs). We characterized the variability of the *Alu*, L1 and SVA families of retrotransposons in different types of sequencing data (whole exome and whole genome sequencing (WES and WGS)) from the PPMI cohort and validated chosen targets using PCR assays in DNA samples from PPMI subjects. This generated genotype data for use in downstream analysis to evaluate the role of RIPs in the development and progression of Parkinson's disease and their impact on gene expression.

Method

Genotyping of retrotransposon insertion polymorphisms in WES and WGS data

The initial screening of WES data for reference RIPs *(Alu*, L1 and SVA families) in exons and their 200bp flanking sequence in 587 exomes from the PPMI cohort was performed using mobile element locator tool deletion (MELT-del) (1). To expand the analysis of these elements genome wide 612 whole genomes from the PPMI cohort were utilized to genotype reference RIPs *(Alu*, L1 and SVA families). This was performed using a combination of the structural variant caller Delly2 (2) and MELT-del (1). In addition to the reference RIPs from the three families a subset of non-reference L1s were also genotyped using MELT-split (1) in order to investigate retrotransposition competent L1s due to their ability to retrotranspose and encode functional proteins.

PCR validation of retrotransposons insertions polymorphism targets in PPMI DNA samples

Depending on the size of the RIP for amplification primers were designed either to amplify the empty site and filled site using a single primer pair or three primers were designed for empty site amplification and either the 5' or 3' junction of the insertion (Table 1). The resulting PCR products were analysed by gel electrophoresis or using the MultiNA (Microchip Electrophoresis System for DNA/RNA Analysis) to genotype each sample for the presence or absence of the insertion based on the product size. The genotypes were recorded as follows: homozygous present (PP), heterozygous (PA) and homozygous absent (AA). All PCR assays, except for SVA_24, were performed using GoTaq G2 hot start polymerase (Promega) under standard conditions. For SVA_24 KOD Hot Start polymerase (Merck) with the addition of 1M betaine was used for amplification. Sanger sequencing of PCR products from selected samples was carried out confirming successful amplification of the correct target.

Table 1: Primers sequences for retrotransposon targets and expected product sizes based on the reference genome.

ID	Forward Primer 5'-3'	Reverse Primer 5'-3'	Internal Primer 5'-3'	Empty site product size (bp)	Filled site/junction product size (bp)
Alu_1393	ATCTCTGTGGCTTCCTGACA	TTTTCTACCACAACGCCAGG	NA	249	594
Alu_1304	TGATTTAGGCTTGTCATTTCCA	TGAACAGGCAGAAGTCTCCA	NA	245	388
Alu_2873	TGTTTGGATCTCAAGAAGCATTT	AAGTGAGTCCCAGGCCTTCT	NA	171	499
SVA_24	TCCCAGGGTCAGAAAAGATG	TAAGCCCCACTGTGATAGGG	NA	304	2062
SVA_27	AGATTTCAGCAATGTCCTCCA	CAACAAACCAGGCACACACT	TGTTTATCTGCTGACCTTCCC	431	423
Ref_RCL1_10	CTGTTCCCTTTGCCTATCCA	TGGCATACTGAGGGCAATTT	AACTCCCTGACCCCTTGC	343	369
Ref_RCL1_14	ATCCTTGCACCGATTCCTCA	GCTCACAATCCCAAACTGCA	AACTCCCTGACCCCTTGC	600	670
NR_RCL1_21	CGAATGGAAAACACGAAAGA	GCGGAAATGGTTAACTCCAA	AACTCCCTGACCCCTTGC	207	344

References

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2. Rausch T, Zichner T, Schlattl A, Stutz AM, Benes V, Korbel JO. DELLY: structural variant discovery by integrated paired-end and split-read analysis. Bioinformatics. 2012;28(18):i333-i9.

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